The second inequality for idèle classes
The author's algebraic proof

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Source: *Class Field Theory*, v4.03, 6 August 2020

Private reading transcription of Theorem VII.5.1 and the supplied algebraic proof in VII§§5–6. The target and its proof occupy physical PDF pages 221–228 (printed pages 212–219); the supporting VII.4.5–4.7 below are on physical pages 220–221. The complete original work is retained at https://www.jmilne.org/math/CourseNotes/CFT.pdf. Mathematical expressions and the author's argument are retained, with connective prose and layout normalized. This is a transcription of published PDF material, not author-native TeX or a new proof.

Notation and the preceding source results

For a number field K, $\mathbb{I}_K$ is its idèle group and $\mathbf{C}_K = \mathbb{I}_K/K^\times$. The diagonal embedding of $K^\times$ is understood. For a finite set S containing the infinite primes, the author's notation is

$$\mathbb{I}_{K,S} = \prod_{v\in S} K_v^\times \times \prod_{v\notin S} U_v, \qquad U_v = \mathcal{O}_v^\times \quad (v \text{ finite}),$$

and $\mathbb{I}_S$ abbreviates $\mathbb{I}_{K,S}$ when the base field is fixed. $U(S)$ is the group of S-units. A local completion above v is denoted by L_w. Norm maps on idèles and on idèle classes use the same symbol. The absolute values are those of the author's product formula; at a complex place the normalized value is the squared usual complex absolute value, as also seen in the calculation in Proposition VII.6.8 below.

The proof uses local class field theory, Kummer theory, the unit theorem, weak approximation, and the basic cohomological restriction and inflation–restriction sequences. The author's already established Theorem VII.4.3 says that $h(\mathbf{C}_L) = [L : K]$ for a finite cyclic extension. Its immediate Corollary VII.4.4, the first inequality, is

$$(\mathbb{I}_K : K^\times \operatorname{Nm}_{L/K} \mathbb{I}_L) \geq [L : K].$$

Indeed, if $h(\mathbf{C}_L) = [L : K]$, its numerator is at least $[L : K]$. The author also uses that $\operatorname{Nm}_{L/K} \mathbb{I}_L$ is open in $\mathbb{I}_K$ (VII.2.8). These are preceding source results, not additional counted targets.

Lemma VII.4.5. *Let L be a finite solvable extension of K (that is, a finite Galois extension with solvable Galois group). If there is a subgroup D of $\mathbb{I}_K$ such that*

(a) $D \subset \operatorname{Nm}_{L/K} \mathbb{I}_L$, and

(b) $K^\times D$ is dense in $\mathbb{I}_K$,

then $L = K$.

Proof. If $L \neq K$, then there is a subfield K' of L that is cyclic over K and different from K. Then

$$D \subset \operatorname{Nm}_{L/K}(\mathbb{I}_L) = \operatorname{Nm}_{K'/K}(\operatorname{Nm}_{L/K}(\mathbb{I}_L)) \subset \operatorname{Nm}_{K'/K}(\mathbb{I}_{K'}).$$

[1] Therefore $K^\times \operatorname{Nm}_{K'/K} \mathbb{I}_{K'}$ is dense in $\mathbb{I}_K$. As it is a union of translates of $\operatorname{Nm}_{K'/K} \mathbb{I}_{K'}$, it is open by VII.2.8; being a subgroup of $\mathbb{I}_K$, it is also closed. Hence $K^\times \operatorname{Nm}_{K'/K} \mathbb{I}_{K'} = \mathbb{I}_K$, and the first inequality implies $K' = K$. □

Proposition VII.4.6. *Let L be a finite solvable extension of K. If $L \neq K$, there are infinitely many primes of K that do not split completely in L.*

Proof. Suppose there are only finitely many, and let $S \supset S_\infty$ be a finite set of primes of K including all those that do not split completely in L. Apply the lemma with

$$D = \mathbb{I}^S \stackrel{\text{def}}{=} \{(a_v) \mid a_v = 1 \text{ for all } v \in S\}.$$

For $w \mid v \notin S$, $L_w = K_v$, so $D \subset \operatorname{Nm}(\mathbb{I}_L)$. Let $\mathbf{a} = (a_v) \in \mathbb{I}_K$. By weak approximation (ANT, 7.20), there is $b \in K^\times$ very close to a_v in K_v for all $v \in S$. Choose $\mathbf{a}' \in \mathbb{I}^S$ whose v-component is that of $b^{-1}\mathbf{a}$ for every $v \notin S$. Then $b\mathbf{a}'$ is close to $\mathbf{a}$ in $\mathbb{I}_K$. Thus $K^\times D$ is dense in $\mathbb{I}_K$. □

Proposition VII.4.7. *Let L/K be a finite solvable extension with Galois group G, and let T be a finite set of prime ideals containing those that ramify in L. The Frobenius elements $(\mathfrak{P}, L/K)$ for $\mathfrak{P} \notin T$ generate G.*

Proof. After enlarging T, we may suppose it is stable under G. The subgroup H generated by the Frobenius elements at the $\mathfrak{P} \notin T$ is then normal. Let $E = L^H$. Recall (V, 1.11) that

$$(\mathfrak{P} \cap E, E/K) = (\mathfrak{P}, L/K)|_E,$$

which is the identity map. Therefore every prime $\mathfrak{p}$ of K not lying under a prime of T splits in E, which shows that $E = K$. By the main theorem of Galois theory, $H = G$. □

The target statement

Theorem VII.5.1. *Let L/K be a Galois extension of number fields with Galois group G. Then*

(a) (Second inequality) *The index $(\mathbb{I}_K : K^\times \operatorname{Nm}(\mathbb{I}_L))$ is finite and divides $[L : K]$.*

(b) $H^1(G, \mathbf{C}_L) = 0$.

(c) $H^2(G, \mathbf{C}_L)$ is finite and its order divides $[L : K]$.

The author's proof: reductions

Lemma VII.5.2. *If G is cyclic, statements (a), (b), and (c) are equivalent, and*

$$(\mathbb{I}_K : K^\times \operatorname{Nm}(\mathbb{I}_L)) = (H^2(G, \mathbf{C}_L) : 1) = [L : K].$$

[1] Source notation note: on physical page 220 (printed page 211), the inner norm in this display is printed with the subscript L/K. That published subscript is retained here as an original notation issue.

Proof. Without restriction on G,

$$\mathbb{I}_K/K^\times \operatorname{Nm}_{L/K}(\mathbb{I}_L) \simeq \mathbf{C}_K/\operatorname{Nm}_{L/K}(\mathbf{C}_L) = H^0_T(G, \mathbf{C}_L).$$

If G is cyclic, its cohomology is periodic, so $H^0_T(G, \mathbf{C}_L) \simeq H^2(G, \mathbf{C}_L)$. This makes (a) and (c) equivalent. Theorem VII.4.3 states that $h(\mathbf{C}_L) = [L : K]$, so each of (a) and (c) is equivalent to (b). □

Lemma VII.5.3. *It suffices to prove the theorem when G is a p-group, p prime.*

Proof. Recall (II, 1.33) that if H is the Sylow p-subgroup of G, then for every G-module M the maps

$$\operatorname{Res} : H^r_T(G, M) \longrightarrow H^r_T(H, M)$$

are injective on the p-primary components. Therefore, if the theorem holds for L/L^H, then p does not divide the order of $H^1_T(G, \mathbf{C}_L)$, and the power of p dividing the orders of $H^0_T(G, \mathbf{C}_L)$ and $H^2_T(G, \mathbf{C}_L)$ is less than the power of p dividing $[L : K]$. Applying this for every p gives the lemma. □

Lemma VII.5.4. *It suffices to prove the theorem when G is cyclic of prime order p.*

Proof. After the last lemma, assume G is a p-group. Induct on $(G : 1)$. Because G is a p-group, it has a normal subgroup H of index p (GT, 4.17). Put $K' = L^H$ and consider the exact sequence (II, 1.34)

$$0 \longrightarrow H^1(G/H, \mathbf{C}_{K'}) \xrightarrow{\text{Inf}} H^1(G, \mathbf{C}_L) \xrightarrow{\text{Res}} H^1(H, \mathbf{C}_L).$$

By assumption $H^1(G/H, \mathbf{C}_{K'}) = 0$, and by induction $H^1(H, \mathbf{C}_L) = 0$. Thus $H^1(G, \mathbf{C}_L) = 0$, proving (b). Because $H^1(H, \mathbf{C}_L) = 0$, the sequence

$$0 \longrightarrow H^2(G/H, \mathbf{C}_{K'}) \longrightarrow H^2(G, \mathbf{C}_L) \longrightarrow H^2(H, \mathbf{C}_L)$$

is exact, which proves (c). Finally,

$$\begin{aligned}(\mathbf{C}_K : \operatorname{Nm}_{L/K} \mathbf{C}_L) = (\mathbf{C}_K &: \operatorname{Nm}_{K'/K} \mathbf{C}_{K'}) \\ &\cdot (\operatorname{Nm}_{K'/K} \mathbf{C}_{K'} : \operatorname{Nm}_{L/K} \mathbf{C}_L).\end{aligned}$$

This divides $p[L : K']$ because $\operatorname{Nm}_{K'/K}$ defines a surjection

$$\mathbf{C}_{K'}/\operatorname{Nm}_{L/K'} \mathbf{C}_L \longrightarrow \operatorname{Nm}_{K'/K} \mathbf{C}_{K'}/\operatorname{Nm}_{L/K} \mathbf{C}_L.$$

□

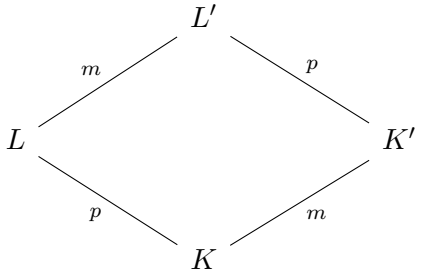
To finish Theorem VII.5.1 using Lemma VII.5.2, it remains to prove the second inequality for cyclic extensions of degree p. The author notes a route through VI.4.9, and then gives the following algebraic proof independent of Chapter VI. The algebraic route is the one retained here.

The algebraic proof of the second inequality

We prove the second inequality when L/K is cyclic of prime degree p.

Lemma VII.6.1. *It suffices to prove the second inequality when K contains a pth root of 1.*

Proof. Let ζ be a primitive pth root of 1 in a fixed algebraic closure of K containing L. Put $K' = K[\zeta]$ and $L' = K'L = L[\zeta]$. Then $[K' : K] = m \mid p-1$, which is relatively prime to p, so $K' \cap L = K$:



The map $\mathrm{Gal}(L'/K) \longrightarrow \mathrm{Gal}(L/K) \times \mathrm{Gal}(K'/K)$ is an isomorphism. Consider the diagram:

$$\begin{array}{ccccccc}
\mathbf{C}_L & \xrightarrow{\mathrm{Nm}_{L/K}} & \mathbf{C}_K & \longrightarrow & \mathbf{C}_K/\mathrm{Nm}\,\mathbf{C}_L & \longrightarrow & 0 \\
\downarrow{\scriptstyle i_L} & & \downarrow{\scriptstyle i_K} & & \downarrow & & \\
\mathbf{C}_{L'} & \xrightarrow{\mathrm{Nm}_{L'/K'}} & \mathbf{C}_{K'} & \longrightarrow & \mathbf{C}_{K'}/\mathrm{Nm}\,\mathbf{C}_{L'} & \longrightarrow & 0 \\
\downarrow{\scriptstyle \mathrm{Nm}_{L'/L}} & & \downarrow{\scriptstyle \mathrm{Nm}_{K'/K}} & & \downarrow & & \\
\mathbf{C}_L & \xrightarrow{\mathrm{Nm}_{L/K}} & \mathbf{C}_K & \longrightarrow & \mathbf{C}_K/\mathrm{Nm}\,\mathbf{C}_L & \longrightarrow & 0
\end{array}$$

Here i_L, i_K are induced by $\mathbb{I}_L \hookrightarrow \mathbb{I}_{L'}$ and $\mathbb{I}_K \hookrightarrow \mathbb{I}_{K'}$. The maps $\mathrm{Nm}_{L/K}$ and $\mathrm{Nm}_{L'/K'}$ are

$$x \longmapsto \sum \sigma x, \qquad \sigma \in \mathrm{Gal}(L/K) \simeq \mathrm{Gal}(L'/K'),$$

and $\mathrm{Nm}_{L'/L}$ and $\mathrm{Nm}_{K'/K}$ are

$$x \longmapsto \sum \sigma x, \qquad \sigma \in \mathrm{Gal}(L'/L) \simeq \mathrm{Gal}(K'/K).$$

The left squares commute, which implies that the rest of the diagram exists. The composites $\mathrm{Nm}_{L'/L} \circ i_L$ and $\mathrm{Nm}_{K'/K} \circ i_K$ are both multiplication by m. Hence the composite

$$\mathbf{C}_K/\mathrm{Nm}\,\mathbf{C}_L \longrightarrow \mathbf{C}_{K'}/\mathrm{Nm}\,\mathbf{C}_{L'} \longrightarrow \mathbf{C}_K/\mathrm{Nm}\,\mathbf{C}_L$$

is also multiplication by m and is therefore an isomorphism: pth powers in $\mathbf{C}_K$ are norms, so $\mathbf{C}_K/\mathrm{Nm}\,\mathbf{C}_L$ is killed by p. In particular, the second map is surjective, and

$$(\mathbf{C}_K : \mathrm{Nm}\,\mathbf{C}_L) \text{ divides } (\mathbf{C}_{K'} : \mathrm{Nm}\,\mathbf{C}_{L'}),$$

which by assumption divides p. □

We prove the second inequality when K contains a primitive pth root of 1 and L is a finite abelian extension of exponent p, with Galois group G. Let $[L : K] = p^r$, so $G \simeq (\mathbb{Z}/p\mathbb{Z})^r$. By Kummer theory (the appendix to Chapter VII), there are $a_1, \ldots, a_r \in K$ such that

$$L = K[a_1^{1/p}, \ldots, a_r^{1/p}].$$

Choose a finite set S of primes of K such that

(a) S contains the infinite primes;

(b) S contains all divisors of p;

(c) all a_i are S-units;

(d) S contains a set of generators for the ideal class group of K, so $\mathbb{I}_K = \mathbb{I}_{K,S}K^\times$ (VII.4.2).

By A.5, (b) and (c) imply that S contains all primes ramifying in L. As usual, $U(S)$ consists of the elements of $K^\times$ that are units at all primes outside S. The unit theorem says

$$U(S) \simeq \mathbb{Z}^{s-1} \times U(S)_{\text{tors}}, \qquad s = |S|.$$

$U(S)_{\text{tors}}$ is cyclic, and its order is divisible by p because it contains a primitive pth root of 1. Therefore

$$U(S)/U(S)^p \simeq (\mathbb{Z}/p\mathbb{Z})^s.$$

Let $M = K[U(S)^{1/p}]$. This is the Kummer extension corresponding to

$$U(S)K^{\times p}/K^{\times p} \simeq U(S)/(U(S) \cap K^{\times p}) = U(S)/U(S)^p \simeq (\mathbb{Z}/p\mathbb{Z})^s.$$

Thus there are extensions

$$M \overset{p^t}{\supset} L \overset{p^r}{\supset} K, \qquad r + t = s.$$

Lemma VII.6.2. *There is a finite set T of primes of K, disjoint from S, such that*

$$\{(\mathfrak{p}_v, M/K) \mid v \in T\}$$

is a basis for $\text{Gal}(M/L)$*, regarded as an $\mathbb{F}_p$-vector space.*

Proof. S contains all primes of K ramifying in M (A.5). For a prime w of M above $v \notin S$, M_w/K_v is therefore unramified, hence cyclic of exponent p. It is either cyclic of order p or trivial. Thus if $M_w \neq L_w$, then $L_w = K_v$; here the author uses w for both the prime of M and the prime of L it divides.

By VII.4.7 there is a finite set T' of primes of M, none above a prime in S, such that $(\mathfrak{p}_w, M/L)$, $w \in T'$, generate $\text{Gal}(M/L)$. After replacing T' by a subset, these Frobenius elements form a basis. Let $w \in T'$ divide w_L in L and w_K in K. Because $M_w \neq L_{w_L}$, we have $L_{w_L} = K_{w_K}$, and so $(\mathfrak{p}_w, M/L) = (\mathfrak{p}_w, M/K)$. Thus $T = \{w_K \mid w \in T'\}$ has the required property. □

The order of T is t, where $p^t = [M : L]$. If w is a prime of M above $v \in T$, then $L_w = K_v$.

Lemma VII.6.3. *With this notation, $a \in U(S)$ becomes a pth power in L if and only if it becomes a pth power in K_v for every $v \in T$.*

Proof. If a becomes a pth power in L, then it does so in L_w for every w. For $w \mid v \in T$, $L_w = K_v$, so it becomes a pth power in K_v. Conversely, let $a \in U(S)$. Then $a^{1/p} \in M$. If $a^{1/p} \in K_v$ for all $v \in T$, it is fixed by $(\mathfrak{p}_v, M/K)$ for all $v \in T$, and hence by $\text{Gal}(M/L)$. Thus $a^{1/p} \in L$ and $a \in L^p$. □

Lemma VII.6.4. *The subgroup*

$$E = \prod_{v \in S} K_v^{\times p} \times \prod_{v \in T} K_v^\times \times \prod_{v \notin S \cup T} U_v$$

of $\mathbb{I}_K$ is contained in $\text{Nm}_{L/K}(\mathbb{I}_L)$.

Proof. Let $\mathbf{a} = (a_v) \in E$. We show that every component a_v is a norm. For $v \in S$, local class field theory gives

$$K_v^\times / \operatorname{Nm} L_w^\times \xrightarrow{\sim} \operatorname{Gal}(L_w/K_v).$$

The second group is killed by p, so the first is also, and $K_v^{\times p} \subset \operatorname{Nm} L_w^\times$. For $v \in T$, $L_w = K_v$, so every element of K_v is a norm from L_w. For $v \notin S \cup T$, L_w/K_v is unramified, so the norm map $U_w \longrightarrow U_v$ is surjective (III, 1.2). □

Now

$$(\mathbf{C}_K : \operatorname{Nm} \mathbf{C}_L) = (\mathbb{I}_K : K^\times \operatorname{Nm} \mathbb{I}_L),$$

which divides $(\mathbb{I}_K : K^\times E)$. It remains to show

$$\boxed{(\mathbb{I}_K : K^\times E) \mid p^r.}$$

But $\mathbb{I}_K = K^\times \mathbb{I}_S = K^\times \mathbb{I}_{S\cup T}$, and hence

$$(\mathbb{I}_K : K^\times E) = (K^\times \mathbb{I}_{S\cup T} : K^\times E).$$

Lemma VII.6.5. *Let A, B, C be subgroups of an abelian group with $A \supset B$. Then*

$$(AC : BC)(A \cap C : B \cap C) = (A : B),$$

in the sense that if two of these indices are finite, then so is the third and the equality holds.

Proof. In the following commutative diagram, the columns and the first two rows are evidently exact. It follows, for example by the snake lemma, that the bottom row is exact, which implies the statement.

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & B \cap C & \longrightarrow & B & \longrightarrow & BC/C & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A \cap C & \longrightarrow & A & \longrightarrow & AC/C & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \dfrac{A \cap C}{B \cap C} & \longrightarrow & A/B & \longrightarrow & AC/BC & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

□

Apply the lemma with $A = \mathbb{I}_{S\cup T}$, $B = E$, $C = K^\times$. Then

$$(\mathbb{I}_K : K^\times E) = \frac{(\mathbb{I}_{S\cup T} : E)}{(U(S \cup T) : K^\times \cap E)}.$$

Lemma VII.6.6. *With the above notation, $(\mathbb{I}_{S\cup T} : E) = p^{2s}$.*

Lemma VII.6.7. *With the above notation, $(U(S \cup T) : K^\times \cap E) = p^{s+t}$.*

Since $r + t = s$, this will prove the boxed formula.

Proof of Lemma VII.6.6. Clearly $(\mathbb{I}_{S\cup T} : E) = \prod_{v\in S}(K_v^\times : K_v^{\times p})$. There are s primes in S, and K contains p distinct pth roots of 1. The next proposition therefore gives

$$(\mathbb{I}_{S\cup T} : E) = \frac{p^{2s}}{\prod_{v\in S}|p|_v}.$$

By assumption S contains every prime at which $|p|_v \neq 1$, so

$$\prod_{v\in S}|p|_v = \prod_{\text{all } v}|p|_v = 1$$

by the product formula. □

Proposition VII.6.8. *Let K be a local field of characteristic zero and let μ_n be the group of nth roots of 1 in K. Then*

$$(K^\times : K^{\times n}) = n\frac{|\mu_n|}{|n|}.$$

If K is nonarchimedean and U is its unit group, then

$$(U : U^n) = \frac{|\mu_n|}{|n|}.$$

Proof. If $K = \mathbb{C}$, the first equation becomes $1 = n \cdot n/n^2$. If $K = \mathbb{R}$, it becomes $1 = n \cdot 1/n$ for odd n and $2 = n \cdot 2/n$ for even n. For nonarchimedean K, $K^\times \simeq U \times \mathbb{Z}$, so

$$(K^\times : K^{\times n}) = (U : U^n)(\mathbb{Z} : n\mathbb{Z}) = (U : U^n)n.$$

It remains to prove the second equation. For an abelian group M, write

$$h_n(M) = \frac{(M : nM)}{(M_n : 1)}, \qquad M_n = \{x \in M \mid nx = 0\}.$$

This is the Herbrand quotient of M regarded as a $\mathbb{Z}/n\mathbb{Z}$-module with trivial action, so the results of II§2 apply. As in the proof of III.2.4, the exponential map defines an isomorphism from a subgroup of finite index in $\mathcal{O}_K$ onto a subgroup of finite index in U. Therefore

$$h_n(U) = h_n(\mathcal{O}_K) = (\mathcal{O}_K : n\mathcal{O}_K) \overset{\text{def}}{=} |n|^{-1}.$$

Hence $(U : U^n) = (U_n : 1)/|n|$. Since $U_n = \mu_n$, the second equation follows. □

Proof of Lemma VII.6.7. Clearly $K^\times \cap E \supset U(S \cup T)^p$. The unit theorem gives

$$(U(S \cup T) : U(S \cup T)^p) = p^{s+t},$$

so it remains to prove $K^\times \cap E \subset U(S \cup T)^p$. The next two lemmas accomplish this; the first shows that the second may be applied to prove the inclusion. □

Lemma VII.6.9. *With the above hypotheses, the obvious map*

$$U(S) \longrightarrow \prod_{v\in T} U_v/U_v^p$$

is surjective.

Proof. Let H be its kernel. To show surjectivity, it is enough to show

$$(U(S) : H) = \prod_{v \in T} (U_v : U_v^p).$$

T is disjoint from S, so $|p|_v = 1$ for $v \in T$. Proposition VII.6.8 shows that the right side is p^t. By Lemma VII.6.3, $H = U(S) \cap L^{\times p}$, whence

$$U(S)/H = U(S)/(U(S) \cap L^{\times p}) \simeq U(S)L^{\times p}/L^{\times p}.$$

By Kummer theory (A.3), the last group corresponds to M/L and therefore has order $[M : L] = p^t$. □

Proposition VII.6.10. *Let K be a number field containing a primitive nth root of 1. Let S be a set of primes containing the infinite primes, the divisors of n, and a set of representatives of the ideal class group of K. Let T be disjoint from S, and suppose that*

$$U(S) \longrightarrow \prod_{v \in T} U_v/U_v^n$$

is surjective. If $b \in K^\times$ is an nth power in K_v for every $v \in S$ and is a unit outside $S \cup T$, then $b \in K^{\times n}$.

Proof. Let $L = K[b^{1/n}]$; we must show that $L = K$. Put

$$D = \prod_{v \in S} K_v^\times \times \prod_{v \in T} U_v^n \times \prod_{v \notin S \cup T} U_v.$$

By Lemma VII.4.5 it suffices to prove (a) $D \subset \mathrm{Nm}_{L/K} \mathbb{I}_L$ and (b) $DK^\times = \mathbb{I}_K$.

(a) Let $\mathbf{d} = (d_v) \in D$. Show that d_v is a norm from $K_v[b^{1/n}]$ for every v. If $v \in S$, then $K_v[b^{1/n}] = K_v$, so every element of K_v is a norm. If $v \in T$, local class field theory says

$$(K_v^\times : \mathrm{Nm}\, K_v[b^{1/n}]^\times) = [K_v[b^{1/n}] : K_v],$$

which divides n, so every nth power in K_v is a norm. If $v \notin S \cup T$, then nb is a unit at v; the field $K_v[b^{1/n}]$ is therefore unramified over K_v, and hence every unit is a norm.

(b) Clearly $\mathbb{I}_S/D = \prod_{v \in T} U_v/U_v^n$, and by hypothesis the map $U(S) \longrightarrow \prod_{v \in T} U_v/U_v^n$ is surjective. Thus $\mathbb{I}_S = D\,U(S)$ and

$$\mathbb{I}_K = \mathbb{I}_S K^\times = D\,U(S)K^\times = DK^\times.$$

□

This completes the proof of Theorem VII.5.1 (the second inequality).

The single collected target is VII.5.1. Its three conclusions, the cyclic and Sylow reductions, the auxiliary Kummer extension and prime set, and VII.6.1–6.10 form one proof group. The preceding first inequality and VII.4.5–4.7 are uncounted source-local support. The source's Remark VII.5.5 and its alternative analytic route through Chapter VI are not additional targets or additional proof routes in this reader.